

GEORGES BANK BUOY 44011, USA

WMO: 992360

Lat: 41.070N

Lon: 66.588W

Elev: 0

StdP: 101.33

Time Zone: -4.00 (W04)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.1	-2.6	-9.9	1.6	-3.6	-8.6	1.8	-2.4	18.9	4.0	17.6	1.4	11.5	280	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.7	23.9	22.8	22.8	22.1	21.9	21.2	23.4	23.7	22.3	22.5	21.4	21.7	6.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.0	23.6	22.2	16.9	22.4	21.3	16.0	21.6	69.4	23.5	65.2	22.4	62.2	21.9	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	15.0	13.7	DB	-5.4	24.1	1.0	1.3	-6.2	25.1	-6.8	25.8	-7.4	26.6	-8.1	27.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	4.5	3.5	4.0	6.2	9.6	13.3	17.6	19.1	17.7	14.1	10.6	7.6	(6)	
	DBStd	6.08	3.78	3.27	2.64	2.00	2.09	2.38	1.99	1.93	2.46	2.85	3.47	3.57	(7)	
	HDD10.0	818	175	183	186	114	32	2	0	0	0	2	32	92	(8)	
	HDD18.3	2867	430	416	444	363	271	153	36	14	40	136	232	333	(9)	
	CDD10.0	1072	4	1	0	1	20	99	237	282	232	128	51	18	(10)	
	CDD18.3	80	0	0	0	0	0	1	15	38	22	4	1	0	(11)	
	CDH23.3	48	0	0	0	0	0	0	2	13	32	2	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	Wind	WSAvg	6.5	8.7	8.7	7.7	6.6	5.3	4.8	4.1	4.5	5.6	6.8	7.5	8.4	(14)
(15) Precipitation	PrecAvg	1689	172	147	141	134	114	108	128	123	120	151	159	192	(15)	
	PrecMax	2005	335	223	241	279	213	253	271	299	206	313	288	296	(16)	
	PrecMin	1284	85	62	64	33	39	12	39	39	24	63	51	125	(17)	
	PrecStd	183	50	44	42	48	42	60	55	53	53	59	46	49	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	14.3	12.0	12.6	18.5	20.6	23.5	24.6	25.5	23.3	21.6	19.3	(19)	
		MCWB	16.4	13.2	12.0	12.4	17.9	20.0	22.3	24.3	24.4	23.3	18.0	18.6	(20)	
	2%	DB	12.8	11.9	10.1	11.2	15.4	19.1	22.1	23.5	24.4	21.2	19.3	17.6	(21)	
		MCWB	12.1	11.1	9.7	10.7	14.5	18.6	21.9	23.2	23.0	19.9	17.5	17.1	(22)	
	5%	DB	11.2	9.7	9.1	10.1	14.1	18.0	21.2	22.7	23.0	19.2	16.6	14.1	(23)	
		MCWB	10.6	9.2	8.6	9.6	13.4	17.5	20.9	22.4	21.1	18.3	15.9	13.7	(24)	
	10%	DB	10.0	7.9	8.0	9.2	12.9	16.8	20.5	22.0	21.5	17.9	15.1	12.6	(25)	
		MCWB	9.2	7.2	7.4	8.7	12.5	16.5	20.2	21.5	19.7	16.9	14.4	11.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	11.7	11.7	11.0	16.8	20.8	22.4	24.2	25.0	20.2	17.4	15.6	(27)	
		MCDB	13.3	12.2	11.8	11.1	17.1	21.0	22.5	24.4	25.2	20.3	17.6	15.7	(28)	
	2%	WB	11.6	9.9	9.4	10.1	14.9	19.1	21.7	23.3	23.9	18.8	16.3	14.5	(29)	
		MCDB	12.0	10.3	9.6	10.2	15.1	19.3	21.7	23.4	24.7	19.1	16.4	14.6	(30)	
	5%	WB	10.5	8.2	8.2	9.2	13.4	18.0	20.9	22.6	21.8	17.8	15.3	13.5	(31)	
		MCDB	11.0	8.7	8.6	9.5	13.6	18.2	21.0	22.7	23.2	18.3	15.5	13.7	(32)	
	10%	WB	9.3	6.6	6.8	8.5	12.3	16.9	20.2	21.8	20.1	16.9	14.4	12.2	(33)	
		MCDB	10.0	7.2	7.2	8.8	12.5	17.1	20.4	21.9	20.8	17.5	14.8	12.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.3	3.6	2.7	3.0	3.0	2.8	2.7	2.8	2.8	3.3	4.2	(35)	
		MCDBR	6.3	7.0	5.6	4.2	4.9	4.2	3.5	3.3	4.0	4.0	4.4	5.5	(36)	
	5% WB	MCWBR	7.3	7.3	6.3	4.7	5.5	4.4	3.4	3.2	3.6	5.1	6.0	6.8	(37)	
		MCWBR	6.3	6.2	5.5	4.3	4.9	4.5	3.5	3.3	3.7	3.8	4.2	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.350	0.349	0.363	0.392	0.417	0.448	0.476	0.434	0.382	0.373	0.352	0.341	(40)	
		taud	2.375	2.409	2.406	2.348	2.319	2.275	2.214	2.338	2.446	2.457	2.457	2.423	(41)	
	Ebn at Noon		790	857	886	880	862	833	805	830	855	820	786	771	(42)	
		Edh at Noon	85	95	106	120	127	133	140	120	101	90	78	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.35	2.16	3.50	4.77	5.47	6.04	5.82	5.50	4.59	3.02	1.79	1.23	(44)	
	RadStd		0.15	0.27	0.32	0.43	0.58	0.49	0.38	0.32	0.29	0.24	0.24	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page